



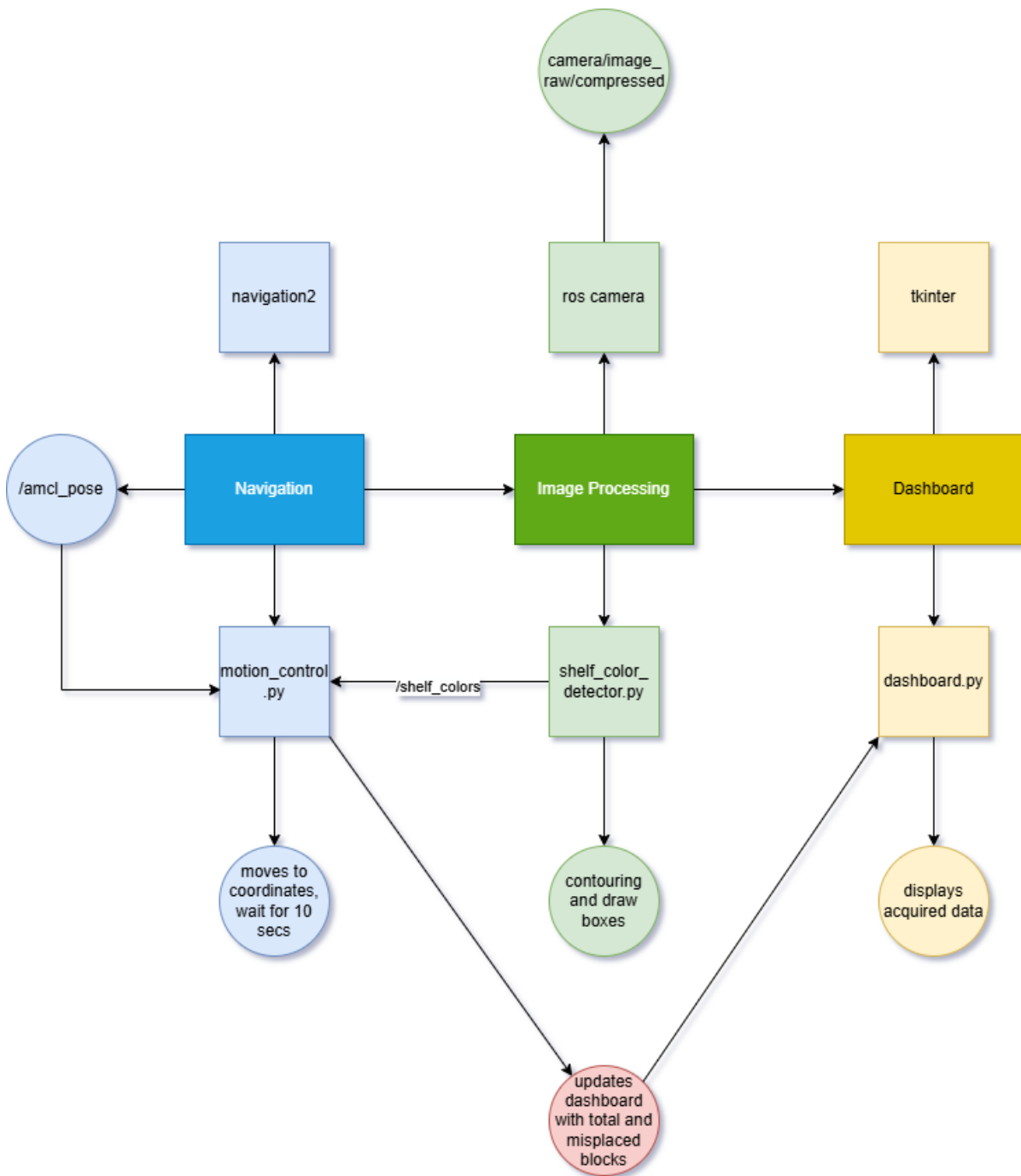
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ECTE477

Project Report

Project Overview

For this project, we used the TurtleBot3 Burger robot running on ROS 2. To control the robot, we connected to it using the command `ssh ubuntu@192.168.15.182 [1]`. We also checked the settings using `nano ~/.bashrc` to make sure the robot and our computer were on the same ID. Once connected, we ran the standard bringup command on the robot itself to turn on the motors and sensors [2]. You can see how these parts work together in the Functional Diagram:



Mapping

In this section, we explored the map and saved the attained map using the following steps:

- Place the bot at the starting position in the maze
- Launch RVIZ [3]
- Launch Cartographer [4]
- Run the Teleop command [5]
- After mapping and confirming the map, save the map to the desired destination using:

```
ros2 run nav2_map_server map_saver_cli -f  
/home/ros/turtlebot3_ws/src/aot_pkg/aot_pkg/maps/map2.yaml [6]
```

- The map can be seen in figure 7 in the appendix.

Navigation

In this section, we had the bot move autonomously, reach three separate points and wait for a while for the image processing. Once complete, the bot returns to the start point. The following steps were used:

- Stop, RVIZ, Cartographer and teleop.
- Run the navigation command:

```
ros2 launch turtlebot3_navigation2 navigation2.launch.py  
map:=/home/ros/turtlebot3_ws/src/aot_pkg/aot_pkg/maps/map2.yaml [8]
```

- This opened RViz.
- To move bot to a point do 2d pose estimate and nav2 goal [9]
- 2d pose estimate was used to define the current orientation (starting point) of the robot on the map.
- Then nav2 goal was used to make the bot move to a specific point on the map and face in a certain direction.
- The AMCL_pose command would print the bot's real time coordinates (the position and orientation). Minor tweaks were then made depending on the current orientation.

```
ros2 topic echo /amcl_pose --once [10]
```

- The coordinates (position and orientation) previously obtained from the amcl_pose topic were hardcoded into the *motion_control.py* script within the self.shelves list. This created a fixed map with targets.

- Instead of manually setting a "nav2 goal" in RViz for every step, It sends the stored coordinates to the action server, instructing the robot to move to the specific shelf autonomously without hitting any walls.
- The navigation feedback is monitored and once the robot successfully reaches the goal, the bot is made to wait for 10 seconds. [11]
- These 10 seconds allow the robot to stabilize completely. During this time, the ROS node stays active so it can receive data and make sure that the Image Processing node has a stable video feed to detect the blocks before the bot moves to the next location.

Image Processing

In this section, we detected the blocks on each shelf and determined their color to count and detect any misplaced blocks. The following steps were used:

- SSH into bot on another window
- Launch camera node on the bot for publishing using:

```
ros2 launch camera_ros camera.launch.py [12]
```

- If first time trying to launch the camera, a command is needed to install:

```
sudo apt install ros-humble-camera-ros [13]
```

- Then on the pc itself, to subscribe to the camera/image_raw/compressed topic, the following installation is made:

```
sudo apt install ros-humble-image-transport-plugins [14]
```

- rqt command is used to view the camera feed on the compressed topic [15]. Images of each shelf are taken here [16].
- The color ranges for each color were then determined by navigating to the util directory in practice_mazes package and accessing the range detector. These were the steps:

- Give permission to use by the command:

```
chmod +x range_detector.py [17]
```

- Use the range detector on the desired picture using the command:

```
./range_detector.py -f HSV -i  
home/ros/turtlebot3_ws/src/aot_pkg/edyelloworangem1.png -p [17]
```

- These ranges were made sure not to overlap to avoid any confusion between different colours using the slider. [18]
- The following are the determined ranges:

- yellow_range = ((26, 73, 190), (35, 255, 255))
 - lime_range = ((43, 79, 0), (50, 145, 255))
 - green_range = ((59, 100, 0), (75, 204, 255))
 - purple_range = ((107, 34, 87), (150, 115, 231))
 - blue_range = ((92, 73, 0), (103, 200, 255))
- Based on these, the colour range values were edited within the *shelf_color_detector.py* code script
 - The code for this image processing part establishes a node that firstly captures the compressed video from the bot's camera, then uses OpenCV to detect & count the blocks based on the defined HSV ranges. To find the blocks, it first performs an opening operation on the image, then finds the contours, and then draws the box around the blocks. It then publishes the block counts to the */shelf_colors* topic for other nodes.
 - *motion_control.py* subscribes to */shelf_colors*, processes the data (calculating totals and misplaced items), and handles the logic.
 - *motion_control.py* then pushes the final processed data into the dashboard by calling the function *self.gui.update_dashboard()*

Dashboard

Tkinter was used to implement the dashboard as a separate Python interface (*dashboard.py*). The updates for the dashboard are taken directly from the *motion_control* node. Three shelf panels with the labels Shelf 1 (Green), Shelf 2 (Purple), and Shelf 3 (Yellow) were shown by the GUI. Every panel displayed:

- Total number of blocks detected
- Number of misplaced blocks
- A list of misplaced colors

The status section showed below the panels:

- The robot's current state ("Moving to Shelf 2", "Scanning Shelf 3", etc.)
- Current robot pose
- Active goal pose
- Total misplaced blocks so far
- Timestamp of the last update

Packages

We created a package called `inventory_control` using the command shown in figure 19 in the appendix. The package contains three different code files:

- `shelf_color_detector`
- `motion_control`
- `dashboard`

Each of these files were executed from the command line using the following script:

```
ros2 run inventory_control <name of file> [20]
```

Results

After all the above steps, we finally ran the bot. The first few attempts the bot successfully stopped at three spots, after waiting 10 seconds at each, but the issue was the bot's orientation was always off by a few units. To fix this, we did a lot of trial and error by manually moving (teleop) the bot to the required points, noting them down and updating the code. The issue still persisted, so we eventually came to the conclusion that it was the network issue and as a result the bot always stopped a few units before the actual goal. So we overcompensated the orientation, w, value by a few units in the code and the bot would orient itself right most of the time.

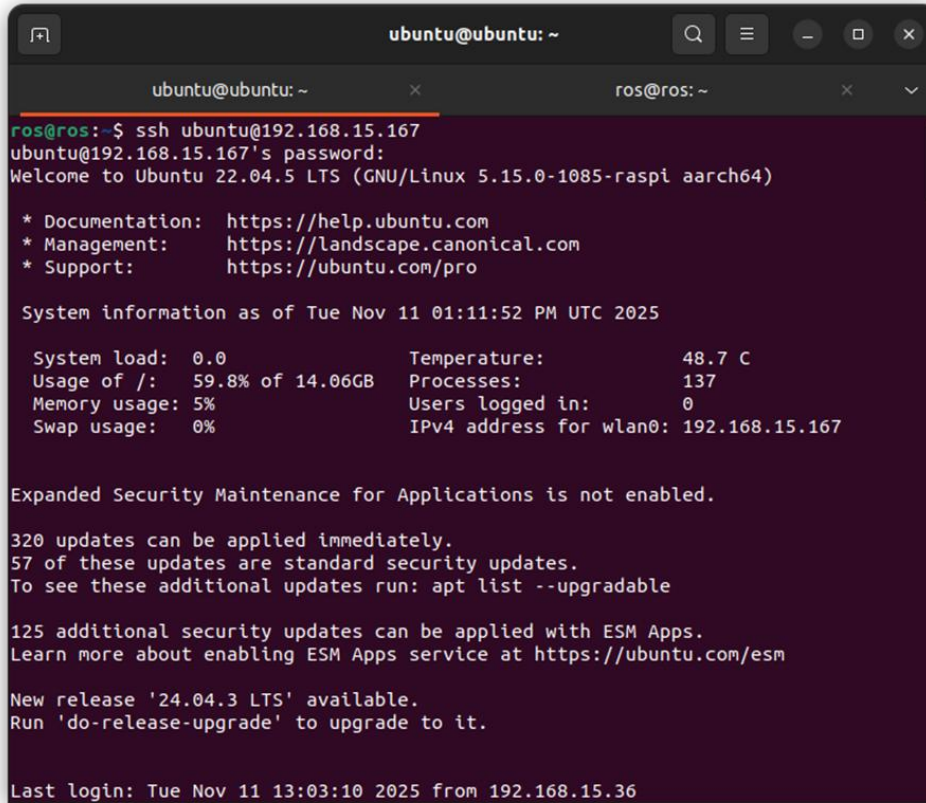
Another issue that we came across was the inaccurate colour detection, we eventually realized that the hsv ranges set for the colours were overlapping with each other. Once we fixed this and made sure the lighting was consistent, the image detection was fixed for good.

After fixing these errors, the bot would run perfectly. It would go to each point, orient itself properly with the camera facing the object, wait for 10 seconds, therefore giving enough time for the image processing and the dashboard to publish the results. Once it finished this for all the colours it would return back to the start point.

Appendix:

Figures:

[1]: CONNECT TO BOT



```
ubuntu@ubuntu: ~
ros@ros: ~
ros@ros:~$ ssh ubuntu@192.168.15.167
ubuntu@192.168.15.167's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-1085-raspi aarch64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Nov 11 01:11:52 PM UTC 2025

System load:  0.0           Temperature:   48.7 C
Usage of /:   59.8% of 14.06GB Processes:    137
Memory usage: 5%           Users logged in: 0
Swap usage:  0%           IPv4 address for wlan0: 192.168.15.167

Expanded Security Maintenance for Applications is not enabled.

320 updates can be applied immediately.
57 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

125 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue Nov 11 13:03:10 2025 from 192.168.15.36
```

[2]: BRINGUP FOR BOT

```
ubuntu@ubuntu: ~  
ubuntu@ubuntu: ~  
ros@ros: ~  
ubuntu@ubuntu:~$ ros2 launch turtlebot3_bringup robot.launch.py  
[INFO] [launch]: All log files can be found below /home/ubuntu/.ros/log/2025-11-11-13-14-08-100162-ubuntu-1636  
[INFO] [launch]: Default logging verbosity is set to INFO  
urdf_file_name : turtlebot3_burger.urdf  
[INFO] [robot_state_publisher-1]: process started with pid [1638]  
[INFO] [ld08_driver-2]: process started with pid [1640]  
[INFO] [turtlebot3_ros-3]: process started with pid [1642]  
[turtlebot3_ros-3] [INFO] [1762866848.884068562] [turtlebot3_node]: Init TurtleBot3 Node Main  
[turtlebot3_ros-3] [INFO] [1762866848.886916117] [turtlebot3_node]: Init DynamixelSDKWrapper  
[robot_state_publisher-1] [INFO] [1762866848.891341636] [robot_state_publisher]: got segment base_footprint  
[robot_state_publisher-1] [INFO] [1762866848.891637006] [robot_state_publisher]: got segment base_link  
[robot_state_publisher-1] [INFO] [1762866848.891693024] [robot_state_publisher]: got segment base_scan  
[robot_state_publisher-1] [INFO] [1762866848.891724858] [robot_state_publisher]: got segment caster_back_link  
[robot_state_publisher-1] [INFO] [1762866848.891753673] [robot_state_publisher]: got segment imu_link  
[robot_state_publisher-1] [INFO] [1762866848.891780080] [robot_state_publisher]: got segment wheel_left_link  
[robot_state_publisher-1] [INFO] [1762866848.891804950] [robot_state_publisher]: got segment wheel_right_link  
[turtlebot3_ros-3] [INFO] [1762866848.895209950] [DynamixelSDKWrapper]: Succeeded to open the port(/dev/ttyACM0)!  
[turtlebot3_ros-3] [INFO] [1762866848.904926580] [DynamixelSDKWrapper]: Succeeded to change the baudrate!
```

[3]: LAUNCH RVIZ

```
ros@ros: ~  
ros@... x ros@... x ros@... x ros@... x ros@... x ros@... x  
ros@ros:~$ ros2 launch nav2_bringup rviz_launch.py  
[INFO] [launch]: All log files can be found below /home/ros/.ros/log/2025-11-11-17-12-10-789215-ros-26278  
[INFO] [launch]: Default logging verbosity is set to INFO  
[INFO] [rviz2-1]: process started with pid [26279]  
[rviz2-1] Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.  
[rviz2-1] [INFO] [1762866731.038907879] [rviz2]: Stereo is NOT SUPPORTED  
[rviz2-1] [INFO] [1762866731.038953619] [rviz2]: OpenGL version: 4.6 (GLSL 4.6)  
[rviz2-1] [INFO] [1762866731.050885712] [rviz2]: Stereo is NOT SUPPORTED  
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT)  
[rviz2-1] [INFO] [1762866781.965650015] [rclcpp]: signal_handler(signum=2)  
[ERROR] [rviz2-1]: process has died [pid 26279, exit code -11, cmd '/opt/ros/humble/lib/rviz2/rviz2 -d /opt/ros/humble/share/nav2_bringup/rviz/nav2_default_view.rviz --ros-args'].  
ros@ros:~$ ros2 launch nav2_bringup rviz_launch.py  
[INFO] [launch]: All log files can be found below /home/ros/.ros/log/2025-11-11-17-13-05-317195-ros-26517  
[INFO] [launch]: Default logging verbosity is set to INFO  
[INFO] [rviz2-1]: process started with pid [26518]  
[rviz2-1] Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.  
[rviz2-1] [INFO] [1762866785.569626651] [rviz2]: Stereo is NOT SUPPORTED  
[rviz2-1] [INFO] [1762866785.569673716] [rviz2]: OpenGL version: 4.6 (GLSL 4.6)
```

[4]: LAUNCH CARTOGRAPHER


```
ros@ros: ~  
ros@... x ros@... x ros@... x ros@... x ros@... x ros@... x  
[cartographer_node-1] [0.675231, 0.700150] Count: 2  
5 (0.694059%) Total: 52 (1.44364%)  
[cartographer_node-1] [0.700150, 0.725069] Count: 3  
1 (0.860633%) Total: 83 (2.30428%)  
[cartographer_node-1] [0.725069, 0.749989] Count: 6  
3 (1.74903%) Total: 146 (4.0533%)  
[cartographer_node-1] [0.749989, 0.774908] Count: 8  
8 (2.44309%) Total: 234 (6.49639%)  
[cartographer_node-1] [0.774908, 0.799827] # Count: 1  
75 (4.85841%) Total: 409 (11.3548%)  
[cartographer_node-1] [0.799827, 0.824747] ## Count: 3  
49 (9.68906%) Total: 758 (21.0439%)  
[cartographer_node-1] [0.824747, 0.849666] ### Count: 5  
64 (15.658%) Total: 1322 (36.7018%)  
[cartographer_node-1] [0.849666, 0.874585] ##### Count: 9  
93 (27.568%) Total: 2315 (64.2699%)  
[cartographer_node-1] [0.874585, 0.899505] ##### Count: 1  
287 (35.7301%) Total: 3602 (100%)  
Optimizing: Done.  
[INFO] [cartographer_occupancy_grid_node-2]: process has finished cleanly [pid 26897]  
[INFO] [rviz2-3]: process has finished cleanly [pid 26899]  
[INFO] [cartographer_node-1]: process has finished cleanly [pid 26895]  
ros@ros:~$ ros2 launch turtlebot3_cartographer cartographer.launch.py
```

[5]: LAUNCH TELEOP

```
ros@ros: ~  
ros@... x ros@... x ros@... x ros@... x ros@... x ros@... x  
ros@ros:~$ ros2 run turtlebot3_teleop teleop_keyboard  
  
Control Your TurtleBot3!  
-----  
Moving around:  
      w  
  a   s   d  
      x  
  
w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffle Pi :  
~ 0.26)  
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waffle Pi  
: ~ 1.82)  
  
space key, s : force stop  
  
CTRL-C to quit  
  
currently: linear velocity 0.01 angular velocity 0.0  
currently: linear velocity 0.02 angular velocity 0.0  
currently: linear velocity 0.0 angular velocity 0.0  
ros@ros:~$ ^C  
ros@ros:~$ ros2 run turtlebot3_teleop teleop_keyboard
```

[6]: SAVE MAP

```
ros@ros: ~  
ros@... x ros@... x ros@... x ros@... x ros@... x ros@... x  
ros@ros:~$ ros2 run nav2_map_server map_saver_cli -f ~/turtlebot3_ws/maps/my_map  
[INFO] [1762867102.713720879] [map_saver]:  
map_saver lifecycle node launched.  
Waiting on external lifecycle transitions to activate  
See https://design.ros2.org/articles/node_lifecycle.html for more inform  
ation.  
[INFO] [1762867102.713844466] [map_saver]: Creating  
[INFO] [1762867102.713924725] [map_saver]: Configuring  
[INFO] [1762867102.714652647] [map_saver]: Saving map from 'map' topic to '/home  
/ros/turtlebot3_ws/maps/my_map' file  
[WARN] [1762867102.714668973] [map_saver]: Free threshold unspecified. Setting i  
t to default value: 0.250000  
[WARN] [1762867102.714677893] [map_saver]: Occupied threshold unspecified. Setti  
ng it to default value: 0.650000  
[WARN] [1762867103.787708020] [map_io]: Image format unspecified. Setting it to:  
pgm  
[INFO] [1762867103.787873646] [map_io]: Received a 59 X 38 map @ 0.05 m/pix  
[INFO] [1762867103.811592891] [map_io]: Writing map occupancy data to /home/ros/  
turtlebot3_ws/maps/my_map.pgm  
[INFO] [1762867103.818120374] [map_io]: Writing map metadata to /home/ros/turtle  
bot3_ws/maps/my_map.yaml  
[INFO] [1762867103.818561446] [map_io]: Map saved  
[INFO] [1762867103.818784671] [map_saver]: Map saved successfully  
[INFO] [1762867103.821786062] [map_saver]: Destroying
```

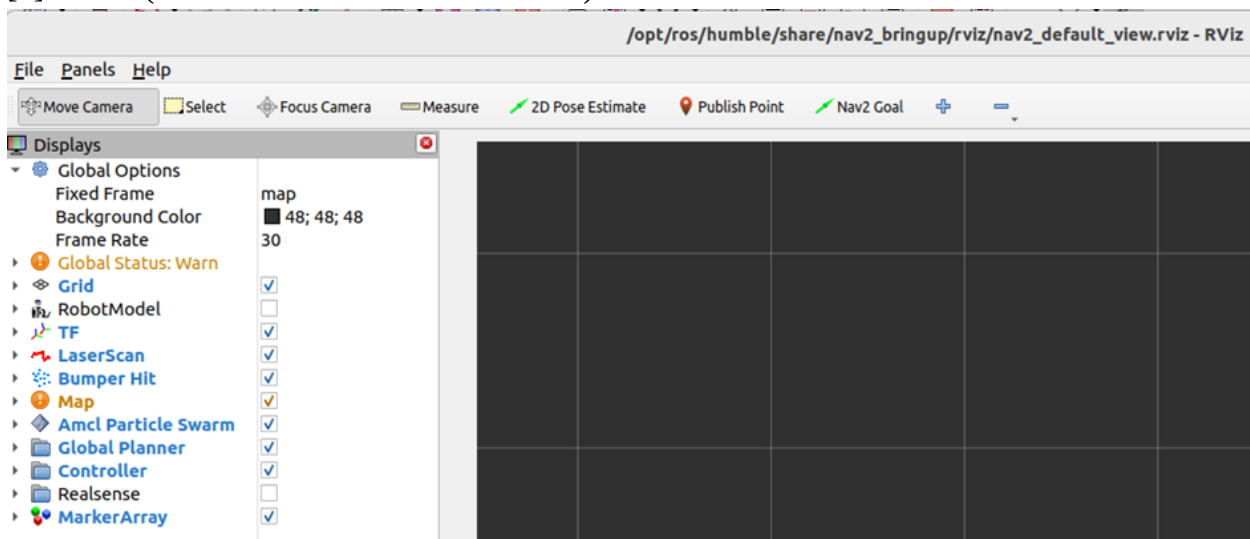
[7]: THE MAP



[8]: NAVIGATION

```
ros@ros: ~  
ble/lib/rviz2/rviz2 -d /opt/ros/humble/share/nav2_bringup/rviz/nav2_default_view  
.rviz --ros-args -r __node:=rviz2 --params-file /tmp/launch_params_k0fcvs6z'].  
^C[WARNING] [launch]: user interrupted with ctrl-c (SIGINT) again, ignoring...  
[component_container_isolated-1] [INFO] [1762868186.835326602] [rclcpp]: signal_  
handler(signum=2)  
[ERROR] [component_container_isolated-1]: process[component_container_isolated-1  
] failed to terminate '5' seconds after receiving 'SIGINT', escalating to 'SIGTE  
RM'  
[INFO] [component_container_isolated-1]: sending signal 'SIGTERM' to process[com  
ponent_container_isolated-1]  
[component_container_isolated-1] [INFO] [1762868190.089564536] [rclcpp]: signal_  
handler(signum=15)  
[ERROR] [component_container_isolated-1]: process[component_container_isolated-1  
] failed to terminate '10.0' seconds after receiving 'SIGTERM', escalating to 'S  
IGKILL'  
[INFO] [component_container_isolated-1]: sending signal 'SIGKILL' to process[com  
ponent_container_isolated-1]  
[ERROR] [component_container_isolated-1]: process has died [pid 27536, exit code  
-9, cmd '/opt/ros/humble/lib/rclcpp_components/component_container_isolated --r  
os-args --log-level info --ros-args -r __node:=nav2_container --params-file /tmp  
/launch_params_itesdq1j --params-file /tmp/launch_params__2j1w8jb -r /tf:=tf -r  
/tf_static:=tf_static'].  
ros@ros: $ ros2 launch turtlebot3_navigation2 navigation2.launch.py map:=/home/e  
ros/turtlebot3_ws/maps/my_map.yaml
```

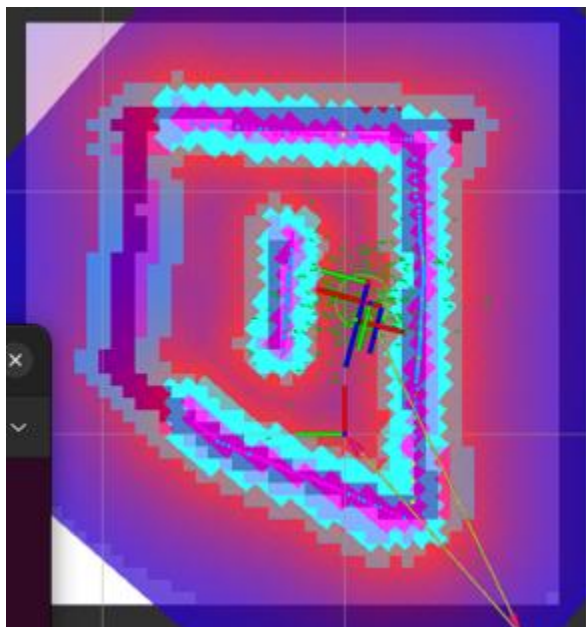
[9]: RVIZ (2D Pose Estimate and Nav2 Goal)



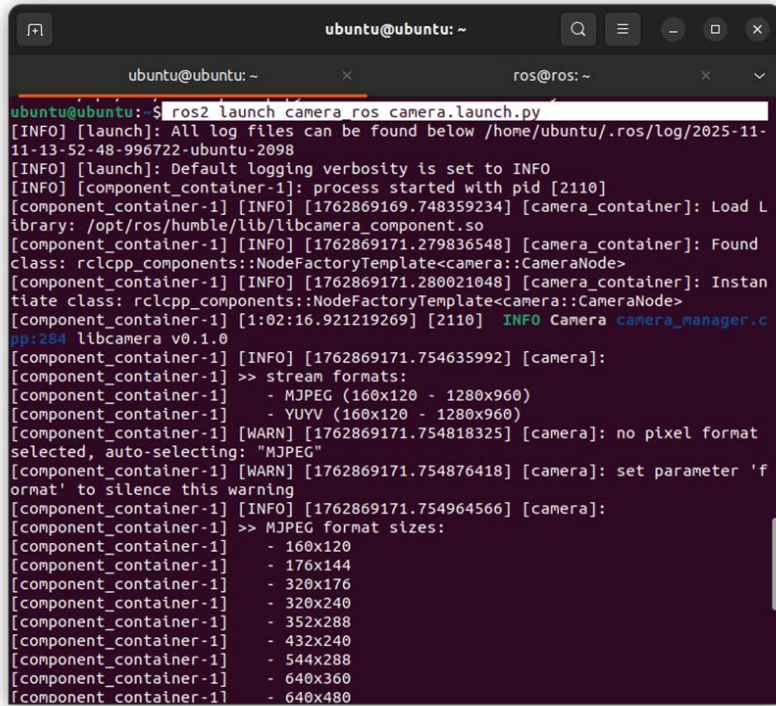
[10]: AMCL

```
ros@ros: ~  
ros@ros: ~  
ros@ros: ~  
ros@ros: ~  
- 0.0  
- 0.0  
- 0.15312325709208263  
...  
ros@ros:~$ ros2 topic echo /ancl_pose --once  
header:  
  stamp:  
    sec: 1763561670  
    nanosec: 962378716  
  frame_id: map  
pose:  
  pose:  
    position:  
      x: 0.5467893060895144  
      y: -0.07463688729823426  
      z: 0.0  
    orientation:  
      x: 0.0  
      y: 0.0  
      z: 0.6217191835836152  
      w: 0.7832402292809805  
  covariance:  
    - 0.02256691022672358  
    - -0.0033329907698981115
```

[11]: WAIT FOR 10 SECONDS



[12]: LAUNCH CAMERA

A terminal window titled 'ubuntu@ubuntu: ~' showing the execution of 'ros2 launch camera ros camera.launch.py'. The output displays various ROS2 log messages, including information about log file locations, logging verbosity, and the successful loading of the camera component. It also shows a list of supported stream formats (MJPEG and YUVV) and a list of MJPEG format sizes. The terminal text is as follows:

```
ubuntu@ubuntu: ~  
ubuntu@ubuntu: ~$ ros2 launch camera ros camera.launch.py  
[INFO] [launch]: All log files can be found below /home/ubuntu/.ros/log/2025-11-11-13-52-48-996722-ubuntu-2098  
[INFO] [launch]: Default logging verbosity is set to INFO  
[INFO] [component_container-1]: process started with pid [2110]  
[component_container-1] [INFO] [1762869169.748359234] [camera_container]: Load L  
ibrary: /opt/ros/humble/lib/libcamera_component.so  
[component_container-1] [INFO] [1762869171.279836548] [camera_container]: Found  
class: rclcpp_components::NodeFactoryTemplate<camera::CameraNode>  
[component_container-1] [INFO] [1762869171.280021048] [camera_container]: Instan  
tiate class: rclcpp_components::NodeFactoryTemplate<camera::CameraNode>  
[component_container-1] [1:02:16.921219269] [2110] INFO Camera camera_manager.c  
pp:284 libcamera v0.1.0  
[component_container-1] [INFO] [1762869171.754635992] [camera]:  
[component_container-1] >> stream formats:  
[component_container-1] - MJPEG (160x120 - 1280x960)  
[component_container-1] - YUVV (160x120 - 1280x960)  
[component_container-1] [WARN] [1762869171.754818325] [camera]: no pixel format  
selected, auto-selecting: "MJPEG"  
[component_container-1] [WARN] [1762869171.754876418] [camera]: set parameter 'f  
ormat' to silence this warning  
[component_container-1] [INFO] [1762869171.754964566] [camera]:  
[component_container-1] >> MJPEG format sizes:  
[component_container-1] - 160x120  
[component_container-1] - 176x144  
[component_container-1] - 320x176  
[component_container-1] - 320x240  
[component_container-1] - 352x288  
[component_container-1] - 432x240  
[component_container-1] - 544x288  
[component_container-1] - 640x360  
[component_container-1] - 640x480
```

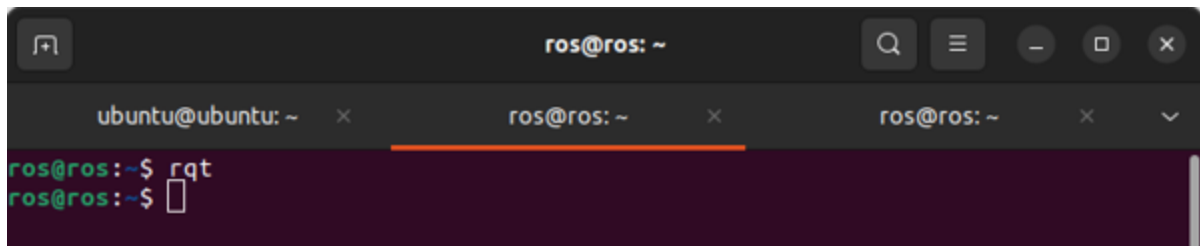
[13]: INSTALL CAMERA

```
ubuntu@ubuntu: ~  
ubuntu@ubuntu: ~  
ubuntu@ubuntu: ~$ sudo apt install ros-humble-camera-ros  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
ros-humble-camera-ros is already the newest version (0.5.0-1jammy.20251010.142317).  
0 upgraded, 0 newly installed, 0 to remove and 325 not upgraded.  
ubuntu@ubuntu:~$ source /opt/ros/humble/setup.py  
-bash: /opt/ros/humble/setup.py: No such file or directory  
ubuntu@ubuntu:~$ ros2 launch camera_ros camera.launch.py  
[INFO] [launch]: All log files can be found below /home/ubuntu/.ros/log/2025-11-11-13-52-48-996722-ubuntu-2098  
[INFO] [launch]: Default logging verbosity is set to INFO  
[INFO] [component_container-1]: process started with pid [2110]  
[component_container-1] [INFO] [1762869169.748359234] [camera_container]: Load Library: /opt/ros/humble/lib/libcamera_component.so  
[component_container-1] [INFO] [1762869171.279836548] [camera_container]: Found Class: rclcpp_components::NodeFactoryTemplate<camera::CameraNode>  
[component_container-1] [INFO] [1762869171.280021048] [camera_container]: Instantiate class: rclcpp_components::NodeFactoryTemplate<camera::CameraNode>  
[component_container-1] [1:02:16.921219269] [2110] INFO Camera camera_manager.cpp:284 libcamera v0.1.0  
[component_container-1] [INFO] [1762869171.754635992] [camera]:  
[component_container-1] >> stream formats:  
[component_container-1] - MJPEG (160x120 - 1280x960)  
[component_container-1] - YUYV (160x120 - 1280x960)  
[component_container-1] [WARN] [1762869171.754818325] [camera]: no pixel format selected, auto-selecting: "MJPEG"  
[component_container-1] [WARN] [1762869171.754876418] [camera]: set parameter 'format' to silence this warning  
[component_container-1] [INFO] [1762869171.754964566] [camera]:  
[component_container-1] >> MJPEG format sizes:
```

[14]:TRANSPORT PLUGINS

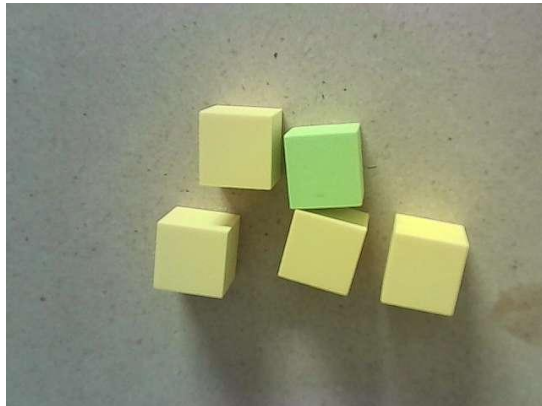
```
ros@ros: ~  
ubuntu@ubuntu: ~  
ros@ros: ~$ sudo apt install ros-humble-image-transport-plugins  
[sudo] password for ros:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following package was automatically installed and is no longer required:  
  libode-dev  
Use 'sudo apt autoremove' to remove it.  
The following additional packages will be installed:  
  ros-humble-compressed-depth-image-transport  
  ros-humble-compressed-image-transport ros-humble-theora-image-transport  
The following NEW packages will be installed:  
  ros-humble-compressed-depth-image-transport  
  ros-humble-compressed-image-transport ros-humble-image-transport-plugins  
  ros-humble-theora-image-transport  
0 upgraded, 4 newly installed, 0 to remove and 327 not upgraded.  
Need to get 471 kB of archives.  
After this operation, 2,533 kB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://packages.ros.org/ros2/ubuntu jammy/main amd64 ros-humble-compressed-depth-image-transport amd64 2.5.4-1jammy.20251017.021653 [121 kB]  
Get:2 http://packages.ros.org/ros2/ubuntu jammy/main amd64 ros-humble-compressed-image-transport amd64 2.5.4-1jammy.20251017.021716 [130 kB]  
Get:3 http://packages.ros.org/ros2/ubuntu jammy/main amd64 ros-humble-theora-image-transport amd64 2.5.4-1jammy.20251017.021814 [213 kB]  
Get:4 http://packages.ros.org/ros2/ubuntu jammy/main amd64 ros-humble-image-transport-plugins amd64 2.5.4-1jammy.20251017.021814 [213 kB]
```

[15]: RQT

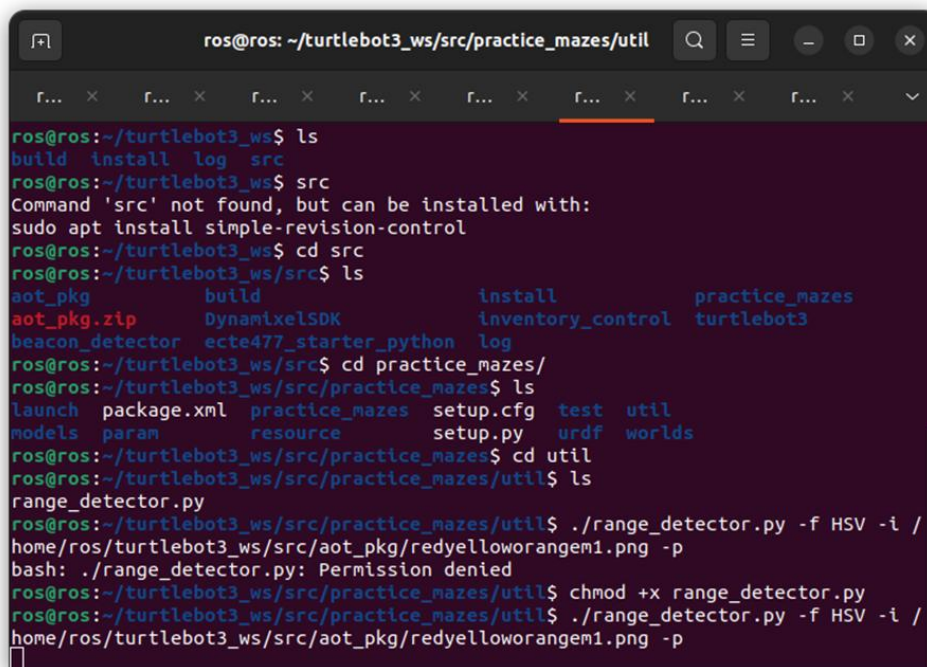


```
ros@ros: ~  
ros@ros:~$ rqt  
ros@ros:~$
```

[16]: BLOCKS EXAMPLE

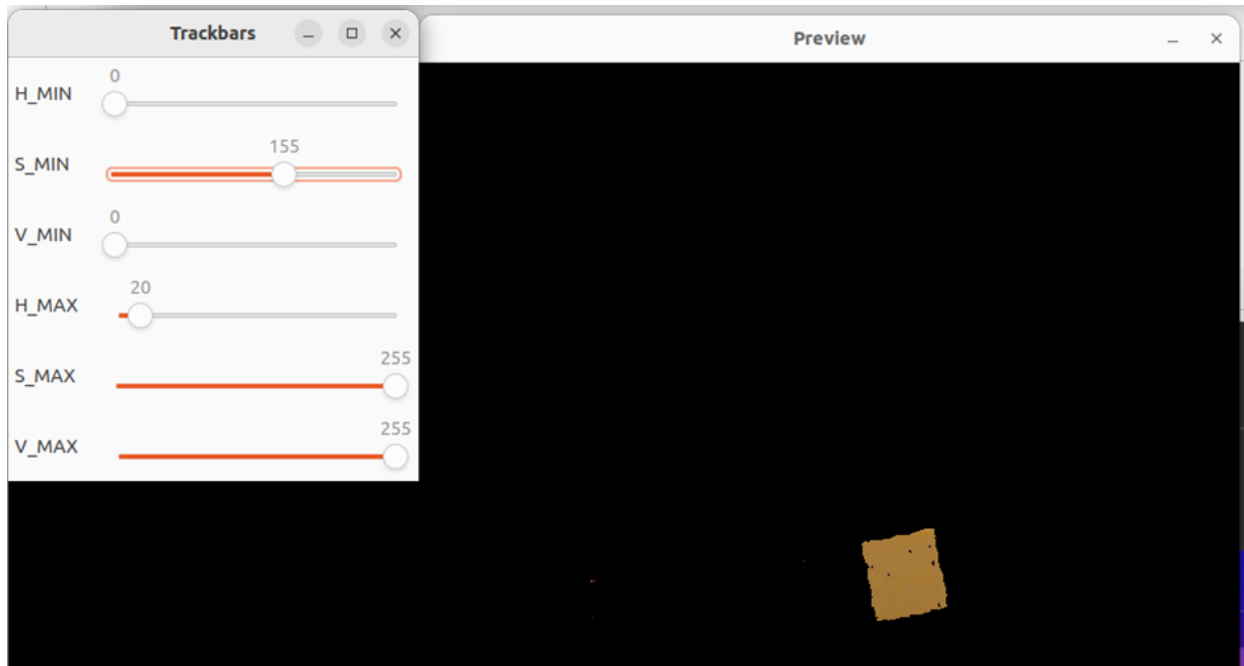


[17]: HSV COLOR RANGES SLIDER



```
ros@ros: ~/turtlebot3_ws/src/practice_mazes/util  
ros@ros:~/turtlebot3_ws$ ls  
build install log src  
ros@ros:~/turtlebot3_ws$ src  
Command 'src' not found, but can be installed with:  
sudo apt install simple-revision-control  
ros@ros:~/turtlebot3_ws$ cd src  
ros@ros:~/turtlebot3_ws/src$ ls  
aot_pkg build install practice_mazes  
aot_pkg.zip DynamixelSDK inventory_control turtlebot3  
beacon_detector ecte477_starter_python log  
ros@ros:~/turtlebot3_ws/src$ cd practice_mazes/  
ros@ros:~/turtlebot3_ws/src/practice_mazes$ ls  
launch package.xml practice_mazes setup.cfg test util  
models param resource setup.py urdf worlds  
ros@ros:~/turtlebot3_ws/src/practice_mazes$ cd util  
ros@ros:~/turtlebot3_ws/src/practice_mazes/util$ ls  
range_detector.py  
ros@ros:~/turtlebot3_ws/src/practice_mazes/util$ ./range_detector.py -f HSV -i /  
home/ros/turtlebot3_ws/src/aot_pkg/redyelloworangem1.png -p  
bash: ./range_detector.py: Permission denied  
ros@ros:~/turtlebot3_ws/src/practice_mazes/util$ chmod +x range_detector.py  
ros@ros:~/turtlebot3_ws/src/practice_mazes/util$ ./range_detector.py -f HSV -i /  
home/ros/turtlebot3_ws/src/aot_pkg/redyelloworangem1.png -p
```

[18]: HSV EXAMPLE



[19]: CREATE INVENTORY CONTROL PACKAGE

```

ros@ros: ~/turtlebot3_ws/src/aot_pkg$ ros2 pkg create --build-type ament_python
inventory_control --dependencies rclpy sensor_msgs cv_bridge std_msgs opencv-py
thon
going to create a new package
package name: inventory_control
destination directory: /home/ros/turtlebot3_ws/src/aot_pkg
package format: 3
version: 0.0.0
description: TODO: Package description
maintainer: ['ros <you@example.com>']
licenses: ['TODO: License declaration']
build type: ament_python
dependencies: ['rclpy', 'sensor_msgs', 'cv_bridge', 'std_msgs', 'opencv-python'
]
creating folder ./inventory_control
creating ./inventory_control/package.xml
creating source folder
creating folder ./inventory_control/inventory_control
creating ./inventory_control/setup.py
creating ./inventory_control/setup.cfg
creating folder ./inventory_control/resource
creating ./inventory_control/resource/inventory_control
creating ./inventory_control/inventory_control/__init__.py
creating folder ./inventory_control/test
creating ./inventory_control/test/test_copyright.py

```

[20]: RUN INVENTORY VISION


```
ros@ros: ~/turtlebot3_ws
Finished <<< aot_pkg [2.30s]
Finished <<< turtlebot3_example [2.33s]
Finished <<< turtlebot3_node [1.37s]
Starting >>> turtlebot3_bringup
Finished <<< turtlebot3_bringup [0.10s]
Starting >>> turtlebot3
Finished <<< dynamixel_sdk_examples [0.17s]
Finished <<< turtlebot3 [0.10s]
Finished <<< ecte477_starter_python [1.81s]
Finished <<< beacon_detector [1.82s]
--- stderr: practice_mazes
/usr/lib/python3/dist-packages/setuptools/dist.py:723: UserWarning: Usage of das
h-separated 'script-dir' will not be supported in future versions. Please use th
e underscore name 'script_dir' instead
  warnings.warn(
---
Finished <<< practice_mazes [1.45s]
Finished <<< inventory_control [2.09s]

Summary: 16 packages finished [3.75s]
  1 package had stderr output: practice_mazes
ros@ros:~/turtlebot3_ws$ source install/setup.bash
ros@ros:~/turtlebot3_ws$ ros2 run inventory_control inventory_vision
```